

Analysis of Connecticut DEEP WQX database Long Island Sound Data

Calder Raio, SUNY ESF

FCH 510, Environmental Chemistry

Final Term Project

May 11, 2026

Write up:

Question 1:

The Long Island Sound (LIS) is a tidal brackish water body between Long Island and Connecticut. It is bordered to the West by the Bronx and Westchester County. To the East it is bordered by the Block Island sound. The northern glacial moraine of Long Island extends beneath the seafloor from Orient Point, LI to Watch Hill, RI. West of the moraine is Long Island Sound, and East of the moraine is Block Island Sound. The LIS is fed by the Atlantic Ocean from the East and feeds back into the Atlantic to the west via the East River (a tidal strait) as it mixes with freshwater from the Hudson River joining the Hudson River estuary known as New York Harbor. Urbanization has appreciably detrimented the Long Island Sound ecosystem since the 70s; eutrophication from sewage and agricultural runoff has created largely anoxic benthic waters, and in 1999, the lobster population nearly completely collapsed, though the precise cause of which is largely not understood due to the lack of aquatic system data particular to the estuary's strata (Bennett et al., 2010). Moreover, *Project Oceanology*, a non-profit ocean science organization, has recorded aquatic system data in the Eastern LIS since 1972. Snyder, et al. (2019) synthesize analyses of this data and reports trends including above-average warming—an increase of 0.45°C per decade in temperature, as well as a rate of acidification twice the global average and an appreciable drop in dissolved oxygen concentrations of 0.29 mgL⁻¹ per decade. This had led to historic declines in marine fauna richness and diversity of cold-water-adapted species like *Homarus americanus*, the American lobster. Their data shows the LIS is not experiencing one-dimensional change, nor is it merely declining in health. Rather, the LIS is a complex, dynamic ecosystem that has experienced historic and difficult to analyse change over the last fifty years. While some cold water economically important species like lobster have seen a decline, according to their data, the common spider crab, *Libinia emarginata*, a native warm-water adapted species, has seen rises in population. These spider crabs may not be as economically important as the lobster or flounder, but they provide ecological benefit as a native predatory decapod.

Question 2:

This write up analyzes data collected from the Connecticut Department of Energy and Environmental Protection agency (CT DEEP) and was accessed via the Water Quality Portal ([WQP](#)). The CT DEEP database comprises records of temperature, pH, pressure, salinity, total dissolved solids (TDS), turbidity, dissolved inorganic carbon (DIC), and alkalinity, as well as chlorophyll, conductivity, dissolved oxygen (DO), DO saturation, Light photosynthetic active radiation at depth (PAR), Phycocyanin. Some of these variables were measured at many depths—these variables were plotted as depth profiles. However, some variables were only measured at surface waters or an unspecified depth. These were recorded as time series, except for alkalinity. However, alkalinity time-series data were used with pH to calculate a predicted time-series DIC, in comparison to the measured DIC time-series. DIC was calculated twice, once assuming ideal conditions, and a second time taking into account both temperature, pressure, density, and ionic strength. For the purposes of this analysis, bottom waters are defined as: 39–45m of depth and surface waters are defined as: <2.05m of depth. Moreover, some of the R scripts include in-text citations, which reference literature bibliographed at the end of this write-up.

Question 3:

Salinity (PSU) and temperature (°C) have seen steady increases between 2008 and 2024, while total dissolved solids (mg/L) and chlorophyll (µg/L) concentrations have decreased over the same time period.

Question 4:

According to the data, based on $[H^+]$ and $[O_{2(aq)}]$, bottom waters are slightly more oxidizing than surface waters. Bottom waters were defined as: 39–45m of depth. Surface waters were defined as: <2.05m of depth. Neither bottom waters nor surface waters have seen any long-term change over time in redox potential. However, the waters become more oxidizing in summer but more reducing in winter. More, the bottom waters seem to lag in response to surface waters. This is reflected in the sinusoidal model wherein bottom waters have a phase shift compared to surface waters. However, both were fit with constant amplitude, so the true difference between bottom and surface waters in redox potential over one year remains yet to be fully understood. Further analyses should include a more accurate sinusoidal model for this data and a more precise analysis of the changes over one year, on average.

Question 5:

Alkalinity and pH were used to calculate $[DIC]$ (ie. C_{T, CO_3}). $[DIC]$ was measured by CT DEEP but the measured values do not match the calculated values. The measured $[DIC]$ falls short of the calculated $[DIC]$. On average, $[DIC]_{calc}$ was 2005.226 µmol/kg and $[DIC]_{measured}$ was 1846.723 µmol/kg, which is a difference of 158.503 µmol/kg.

Question 6:

[DIC] was recalculated assuming a non-ideal system, taking into account the temperature, pressure, and salinity, which account for density and can be used to calculate ionic strength using the Millero equation (Millero & Leung, 1976). The $[\text{DIC}]_{\text{non-ideal}}$ was 1934.683 $\mu\text{mol/kg}$ on average, which is 87.96 $\mu\text{mol/kg}$ higher than $[\text{DIC}]_{\text{measured}}$ but 70.542 $\mu\text{mol/kg}$ lower than $[\text{DIC}]_{\text{ideal}}$ (ie. $[\text{DIC}]_{\text{calc}}$). See the table below for a concise description.

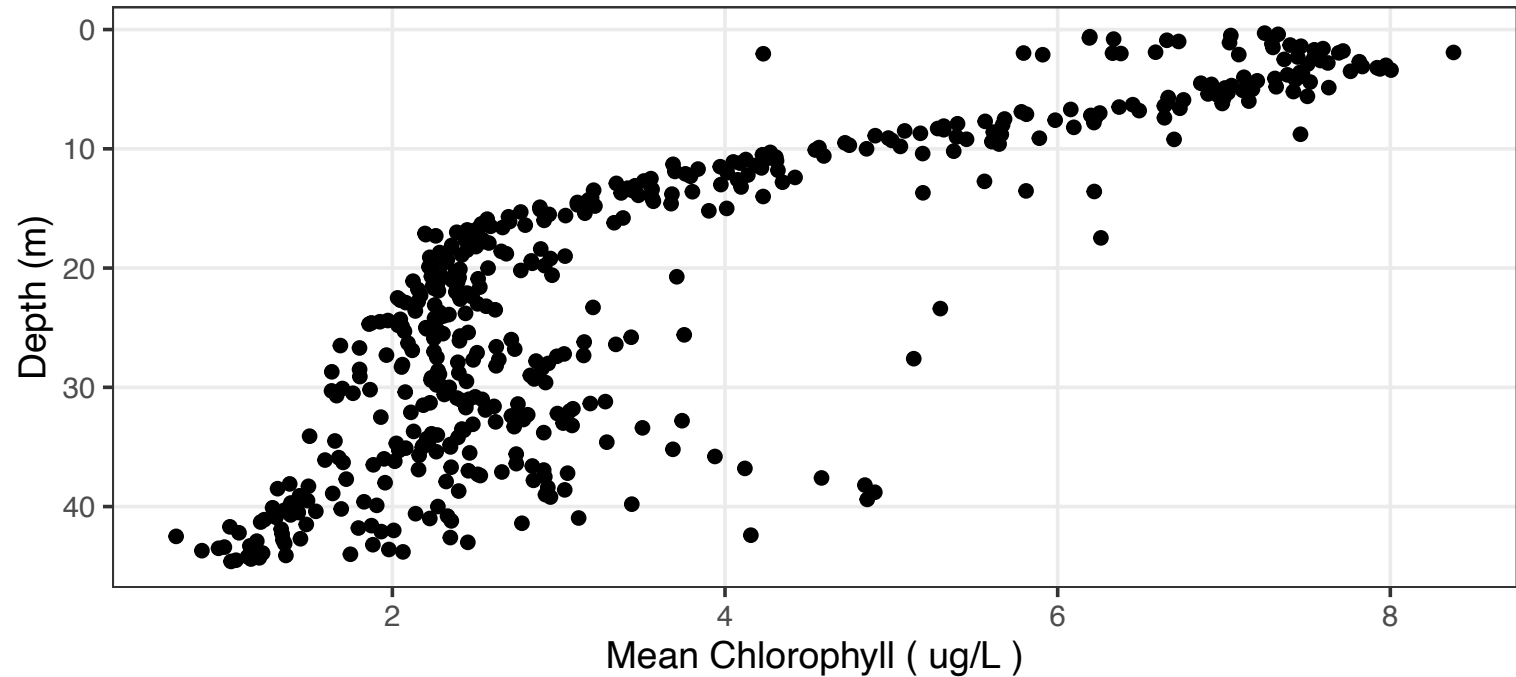
Table 1. Mean [DIC] values calculated from [Alk] and pH versus mean measured [DIC] values.

[DIC] type	Mean [DIC] ($\mu\text{mol/kg}$)
Calculated ideal	2005.226
Calculated nonideal	1934.683
Measured	1846.723

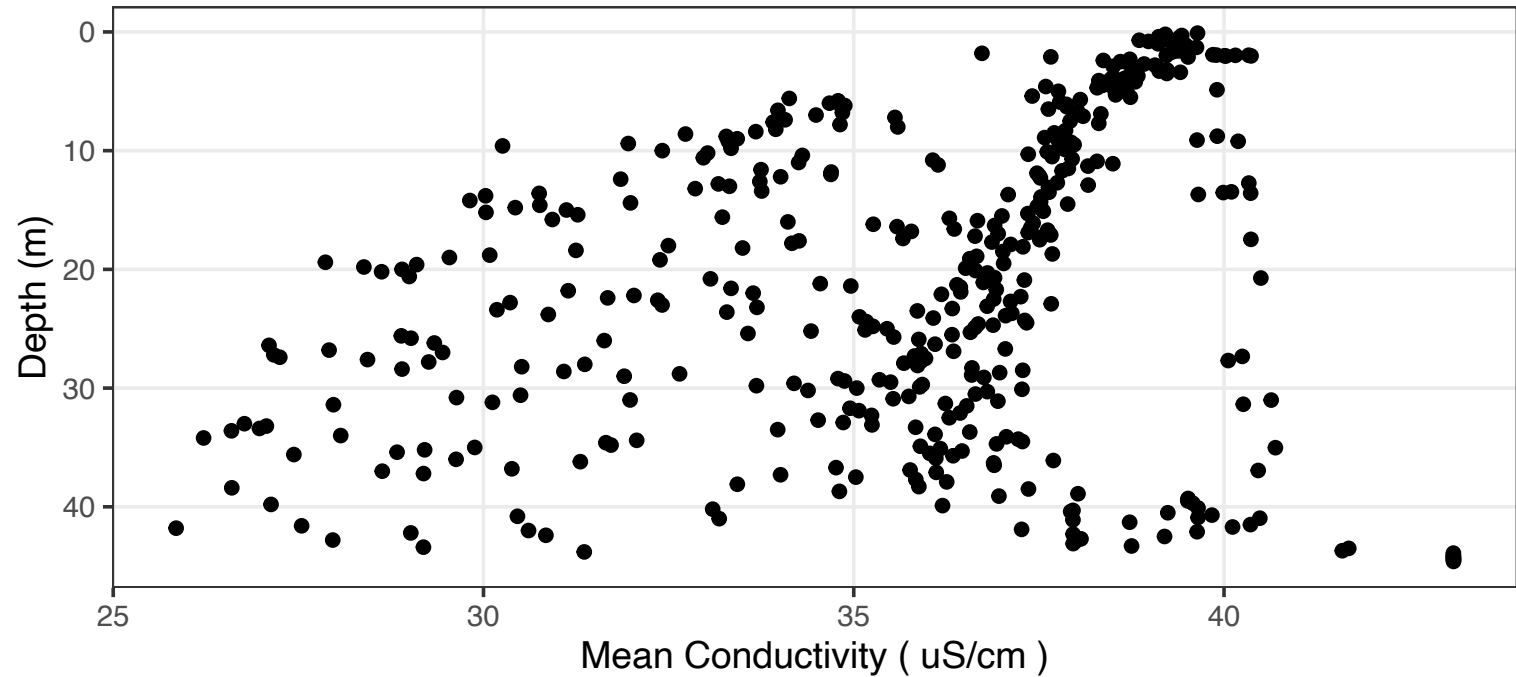
References

- Bennett, D., O'Donnell, J., Bohlen, W., & Houk, A. (2010). Tides and Overtides in Long Island Sound. *Journal of Marine Research*, 68(1).
https://elischolar.library.yale.edu/journal_of_marine_research/257
- Malmberg, C. G., & Maryott, A. A. (1956). Dielectric constant of water from 0 to 100 C. *Journal of Research of the National Bureau of Standards*, 56(1), 1.
<https://doi.org/10.6028/jres.056.001>
- Millero, F. J., & Leung, W. H. (1976). The thermodynamics of seawater at one atmosphere. *American Journal of Science*, 276(9). <https://doi.org/10.2475/ajs.276.9.1035>
- Snyder, J. T., Whitney, M. M., Dam, H. G., Jacobs, M. W., & Baumann, H. (2019). Citizen science observations reveal rapid, multi-decadal ecosystem changes in eastern Long Island Sound. *Marine Environmental Research*, 146, 80–88.
<https://doi.org/10.1016/j.marenvres.2019.03.007>

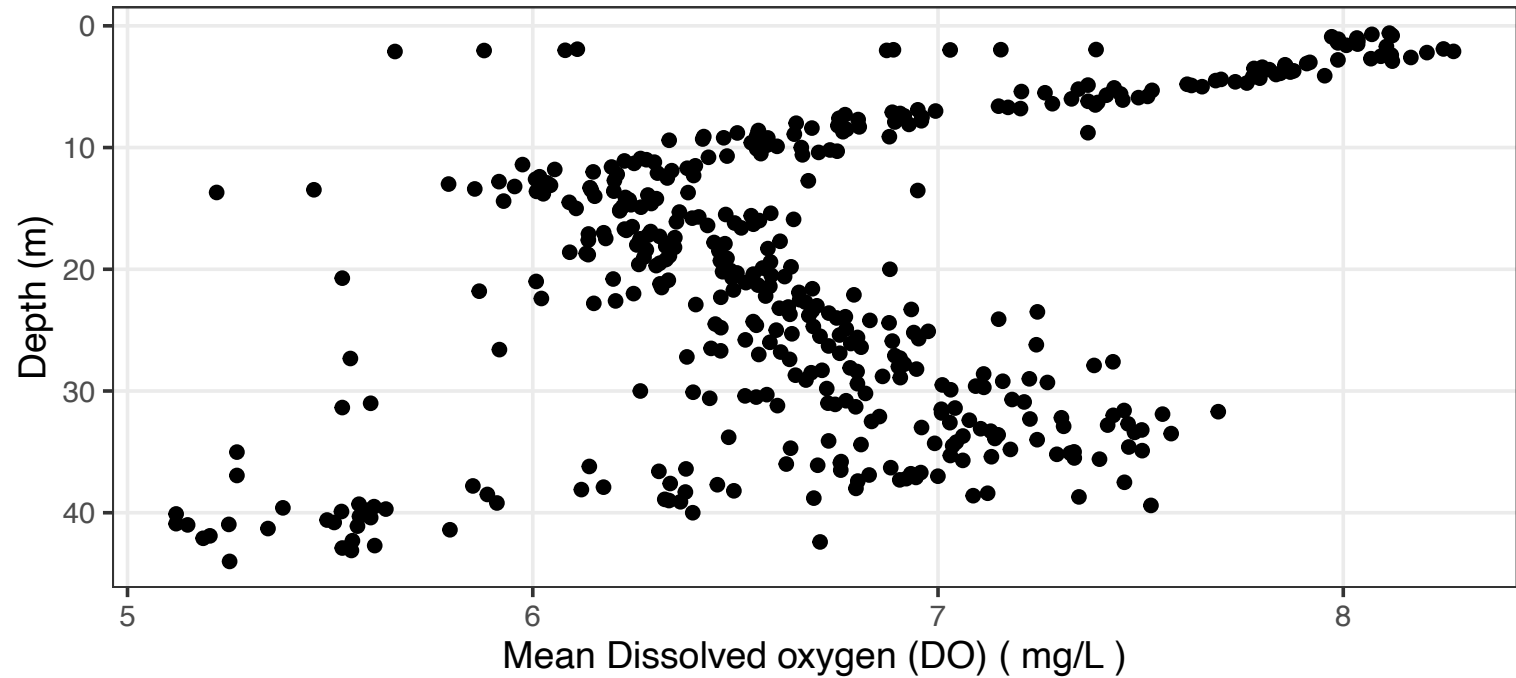
Time-Averaged Depth Profile 2008–2024: Chlorophyll



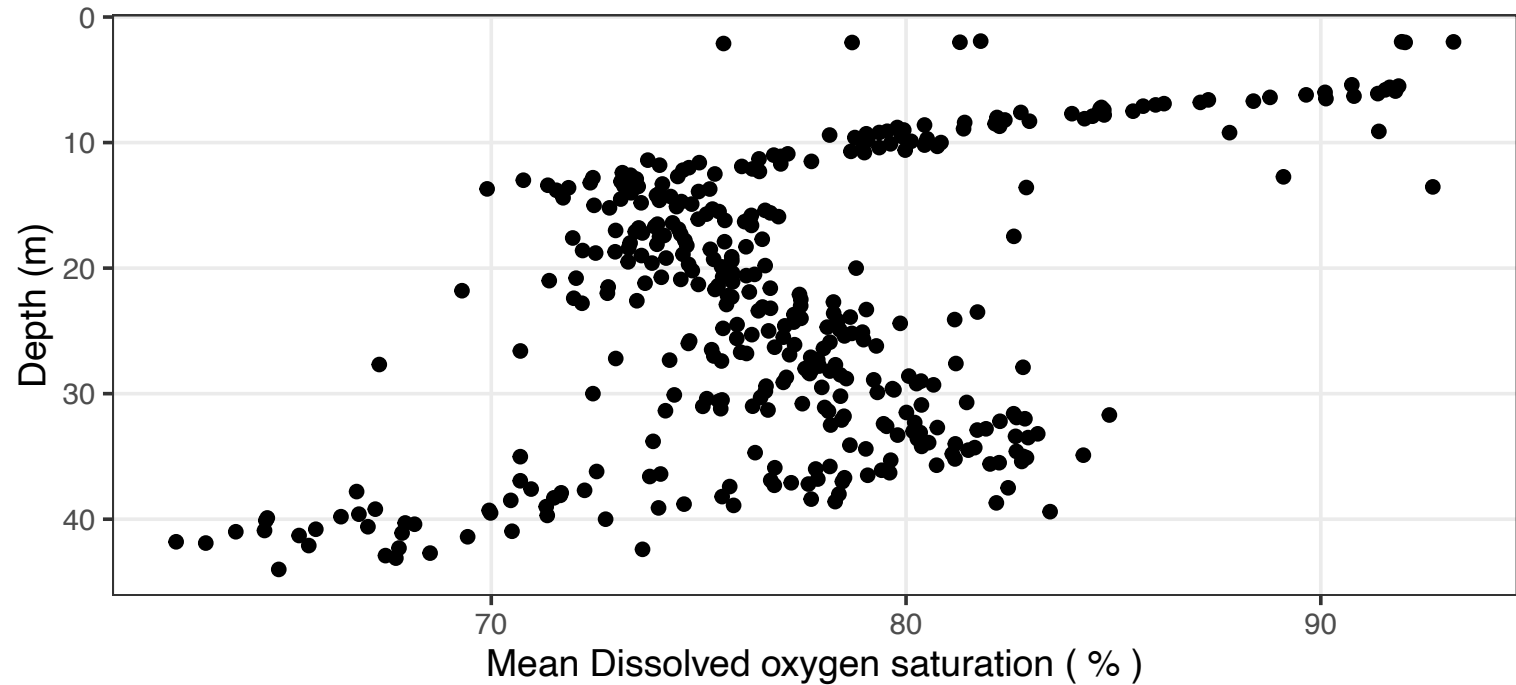
Time-Averaged Depth Profile 2008–2024: Conductivity



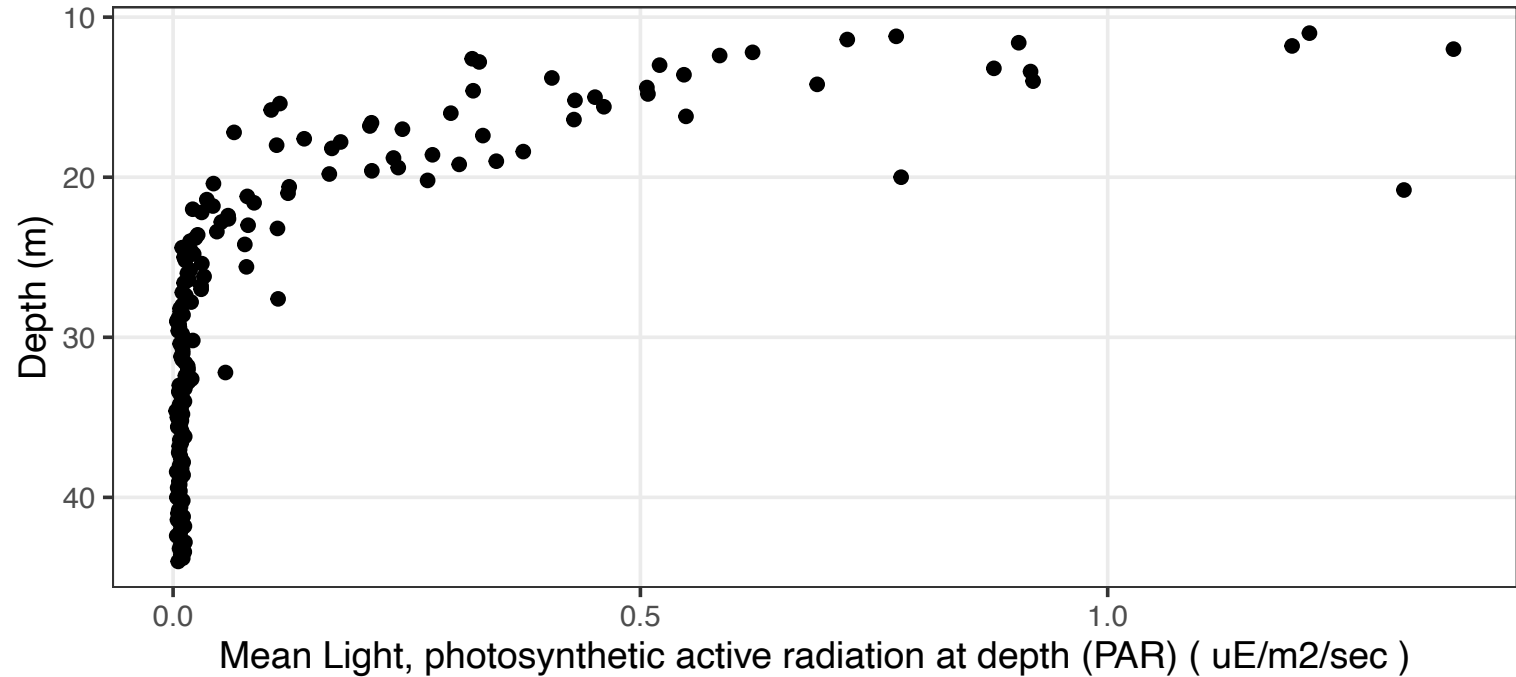
Time-Averaged Depth Profile 2008–2024: Dissolved oxygen (DO)



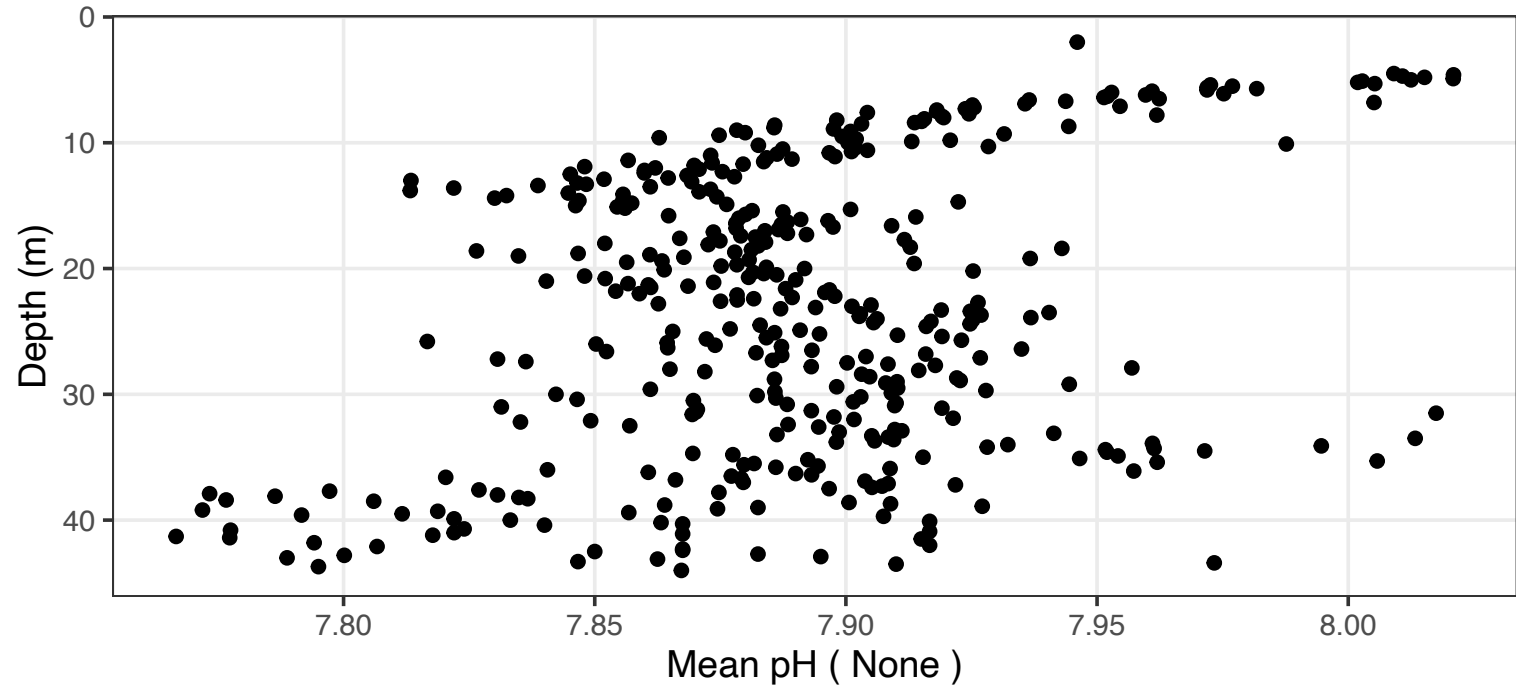
Time-Averaged Depth Profile 2008–2024: Dissolved oxygen saturation



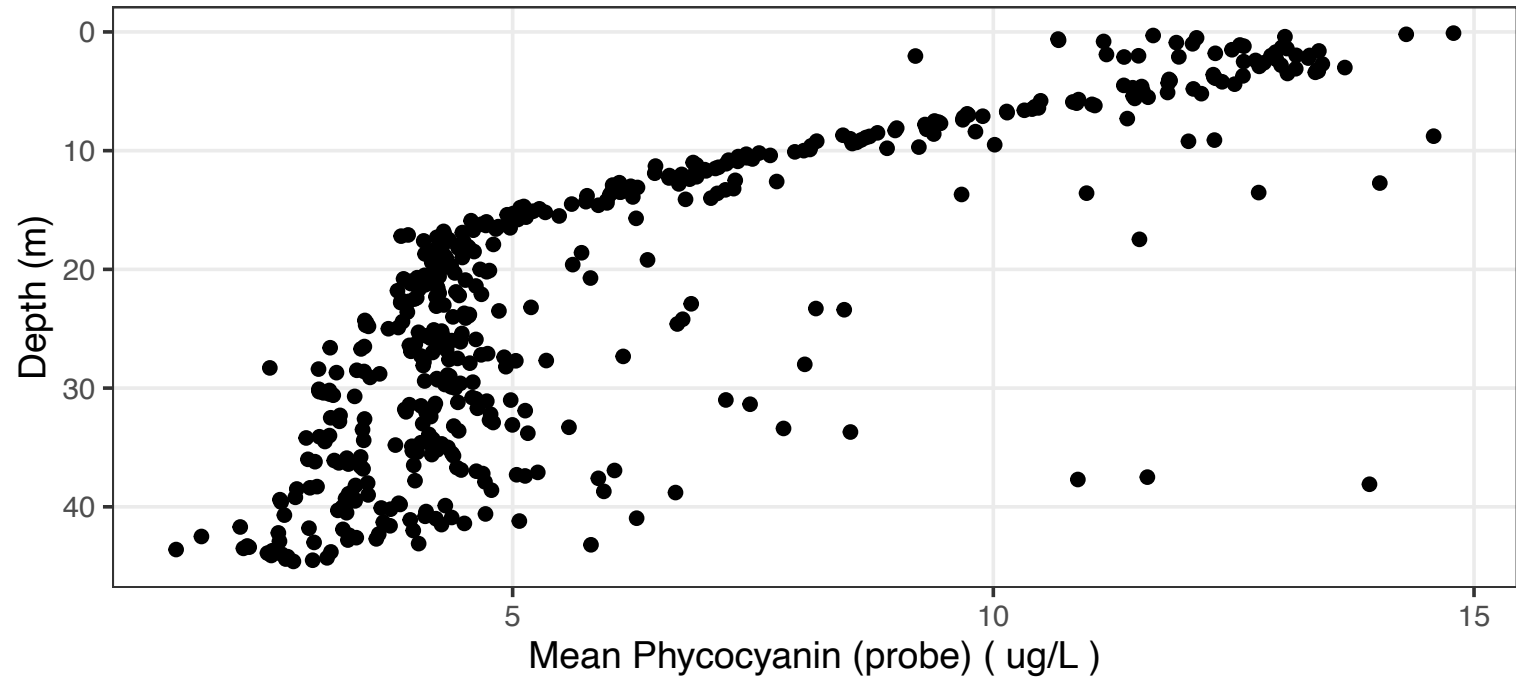
Time-Averaged Depth Profile 2008–2024: Light, photosynthetic ac



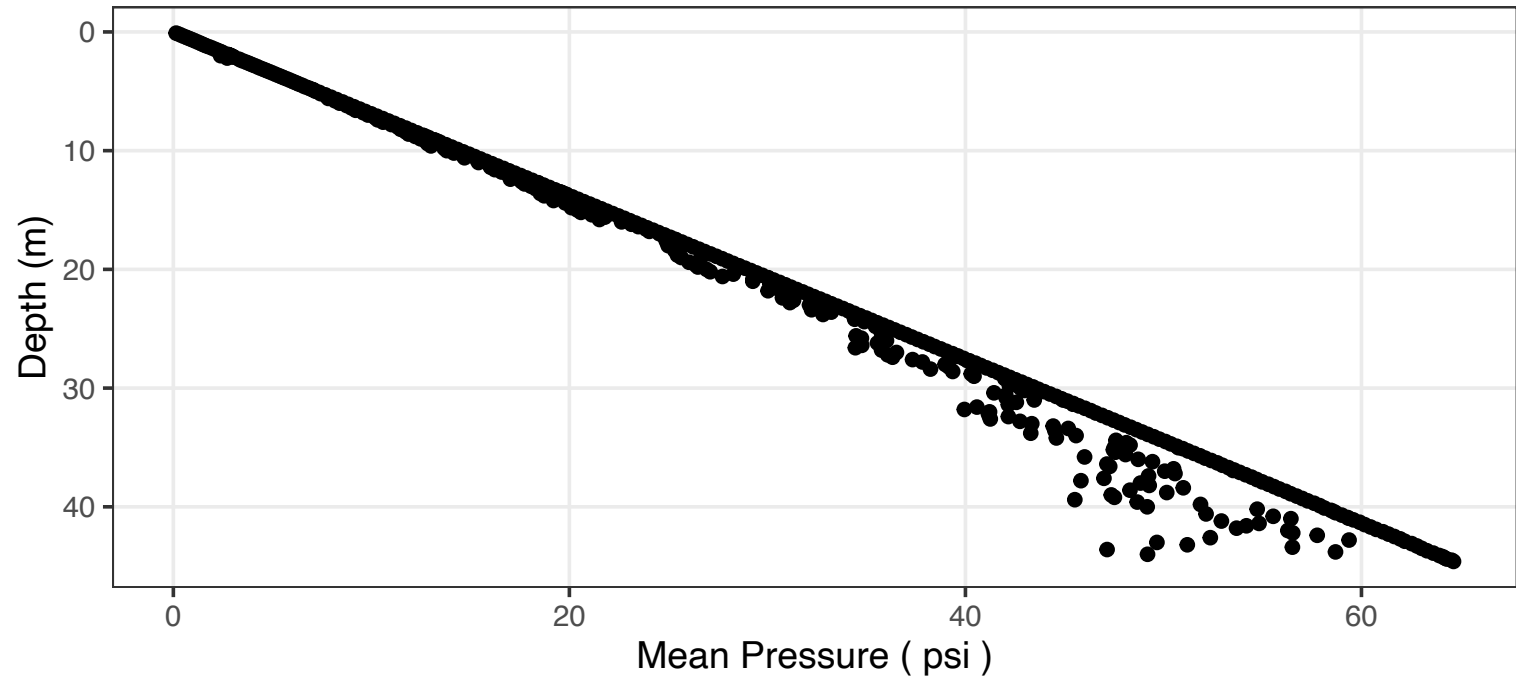
Time-Averaged Depth Profile 2008–2024: pH



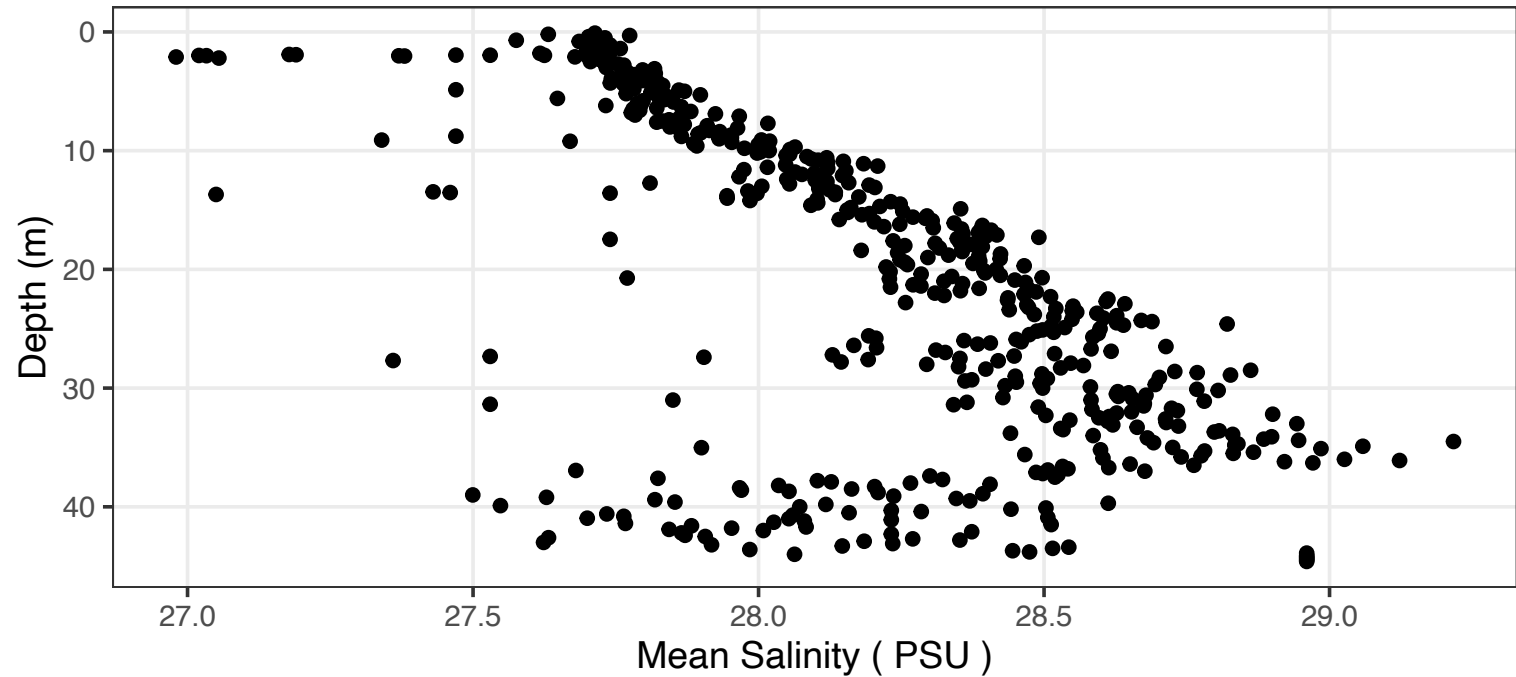
Time-Averaged Depth Profile 2008–2024: Phycocyanin (probe)



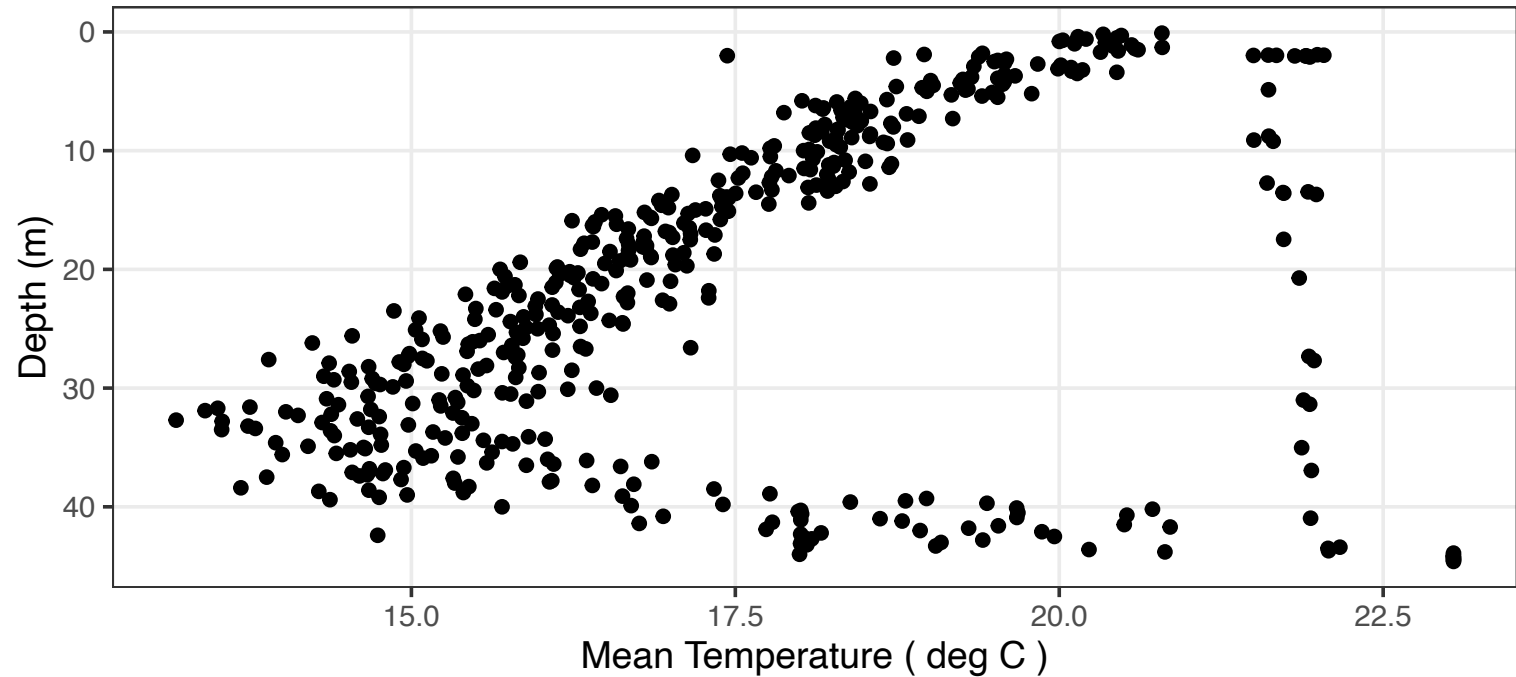
Time-Averaged Depth Profile 2008–2024: Pressure



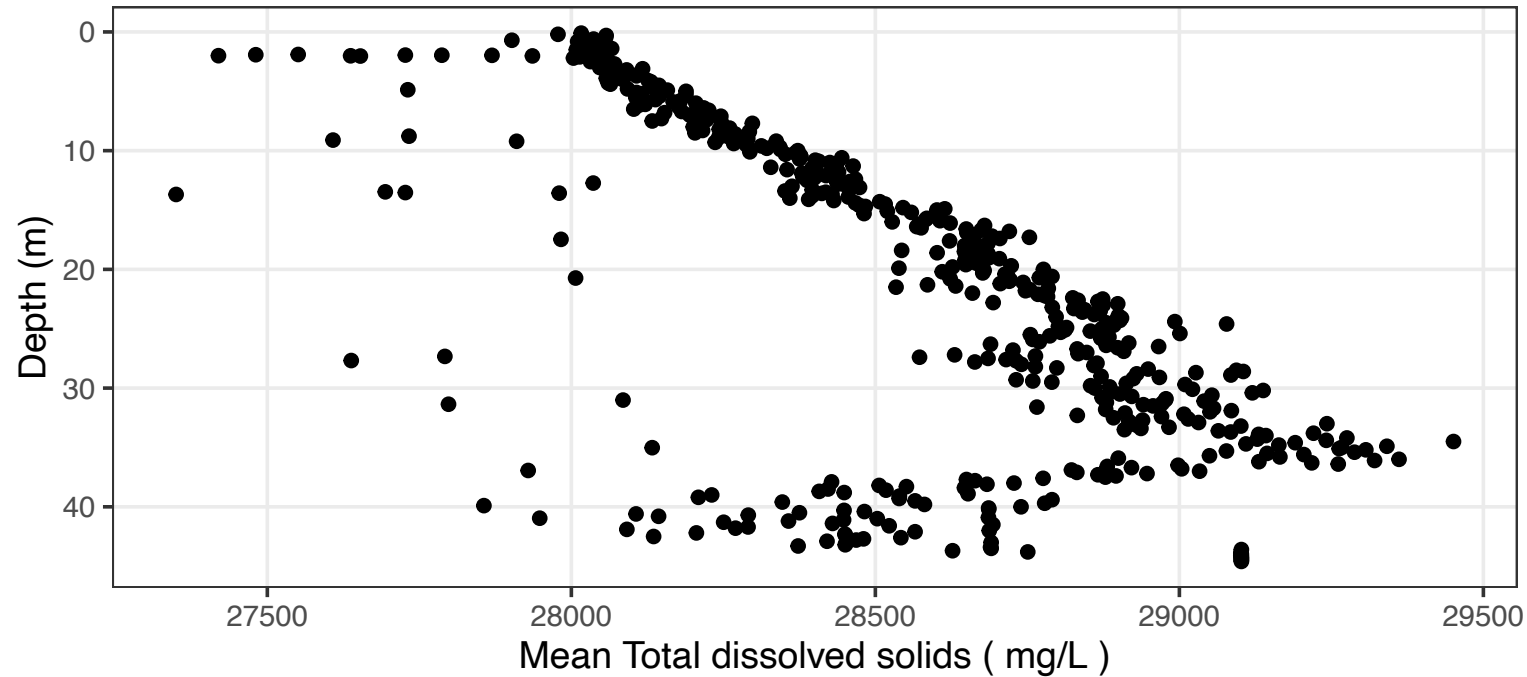
Time-Averaged Depth Profile 2008–2024: Salinity



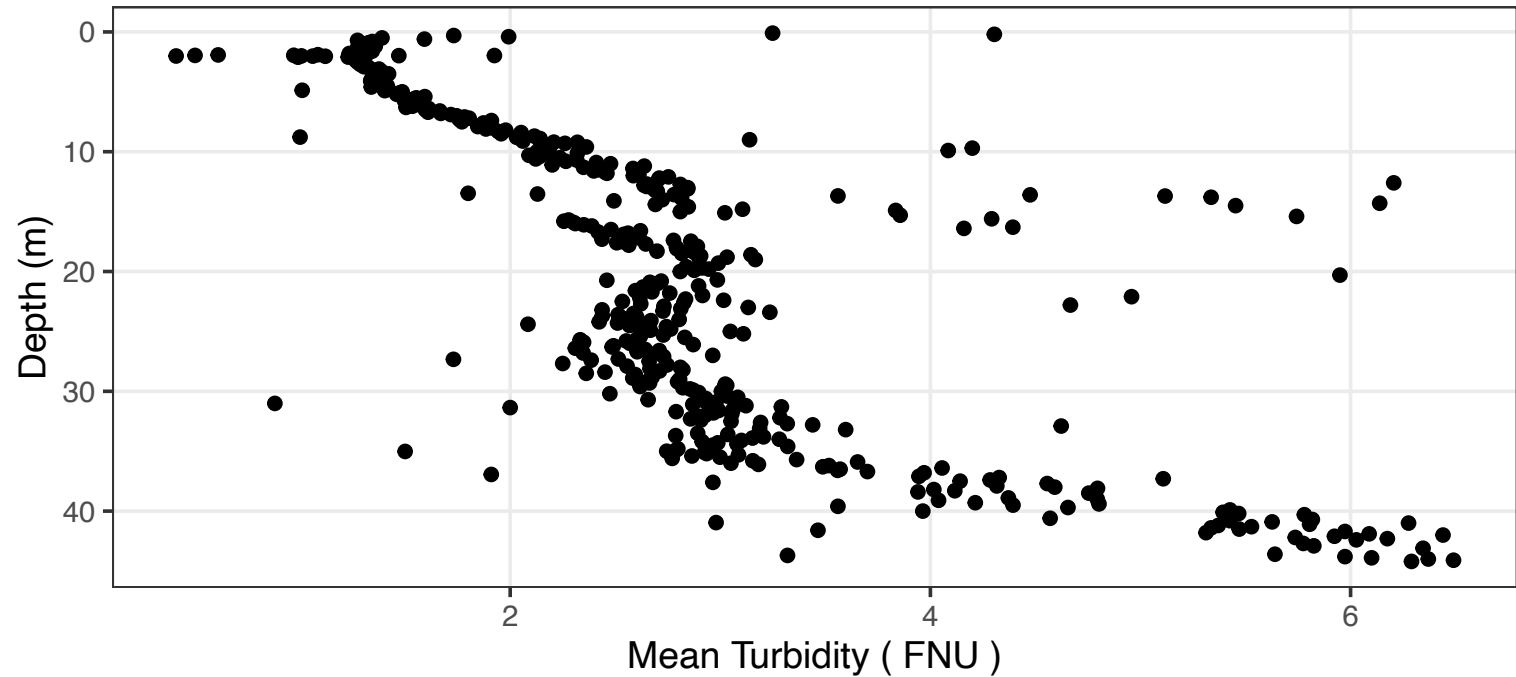
Time-Averaged Depth Profile 2008–2024: Temperature



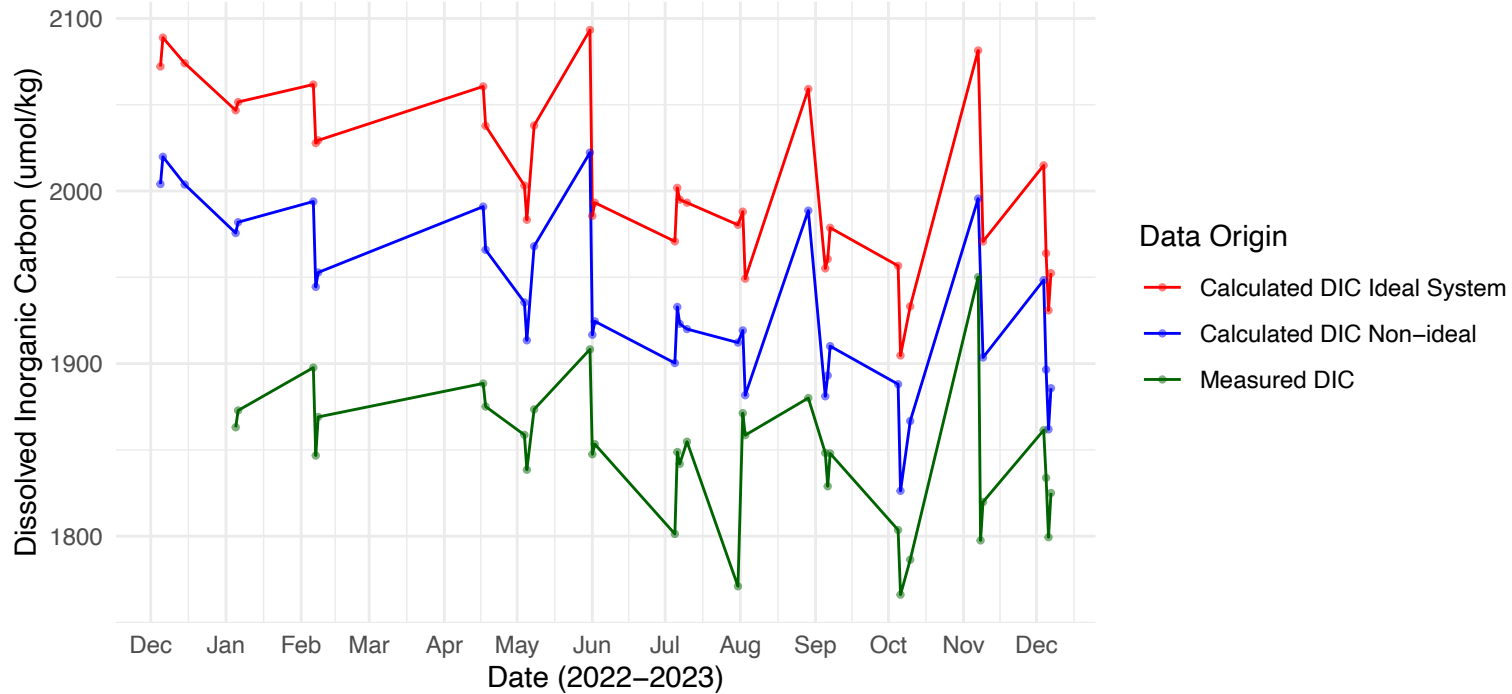
Time-Averaged Depth Profile 2008–2024: Total dissolved solids



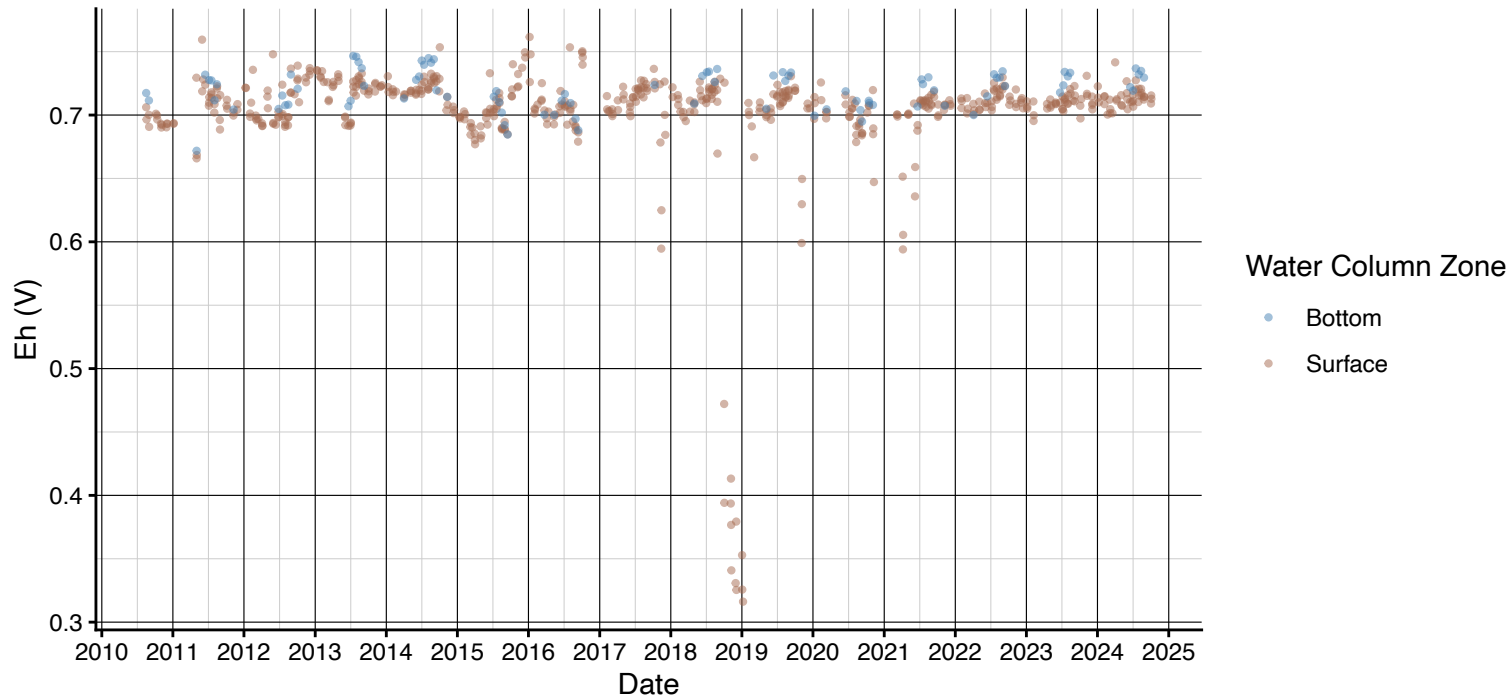
Time-Averaged Depth Profile 2008–2024: Turbidity



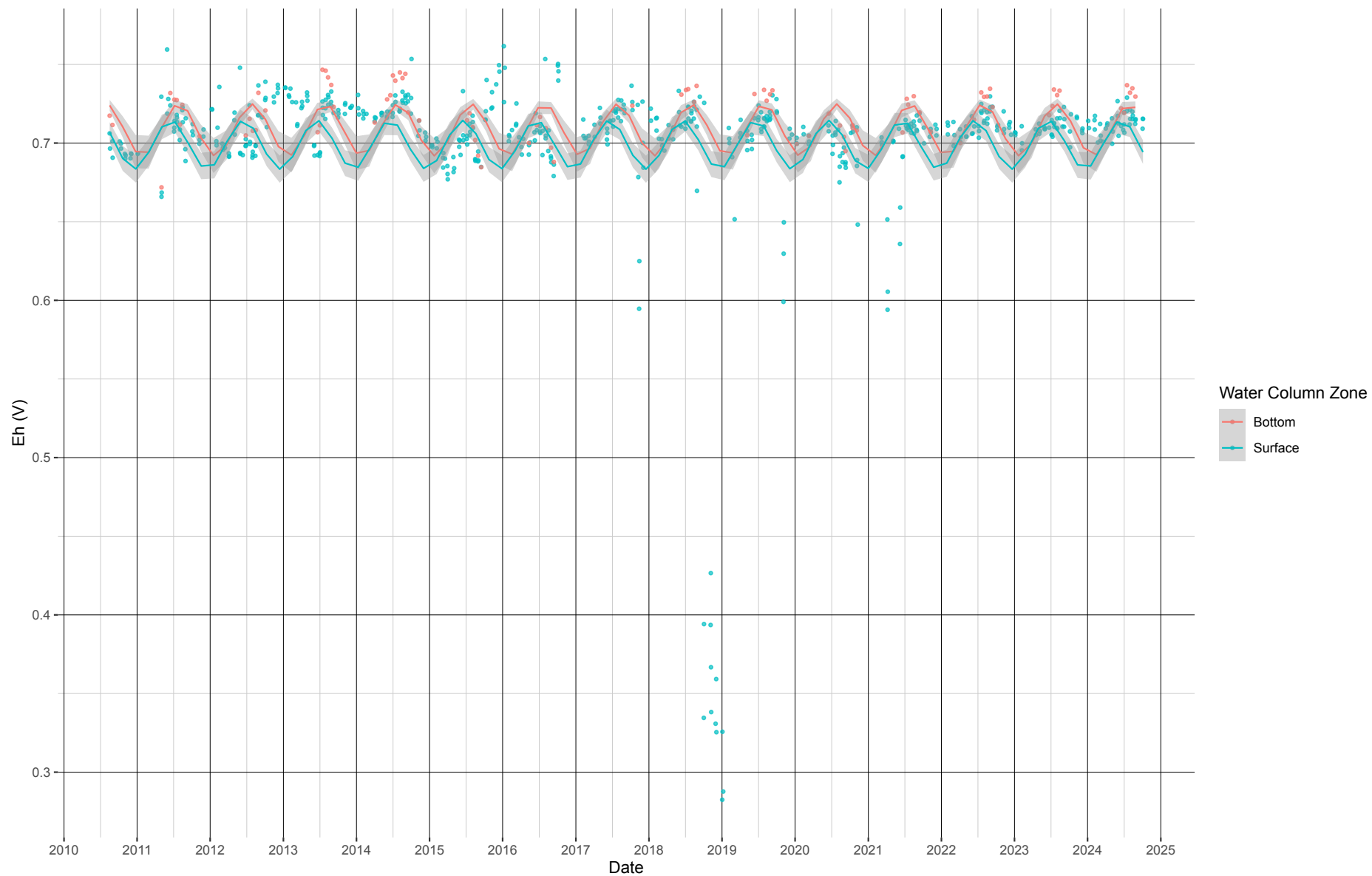
Calculated DIC and Measured DIC versus time



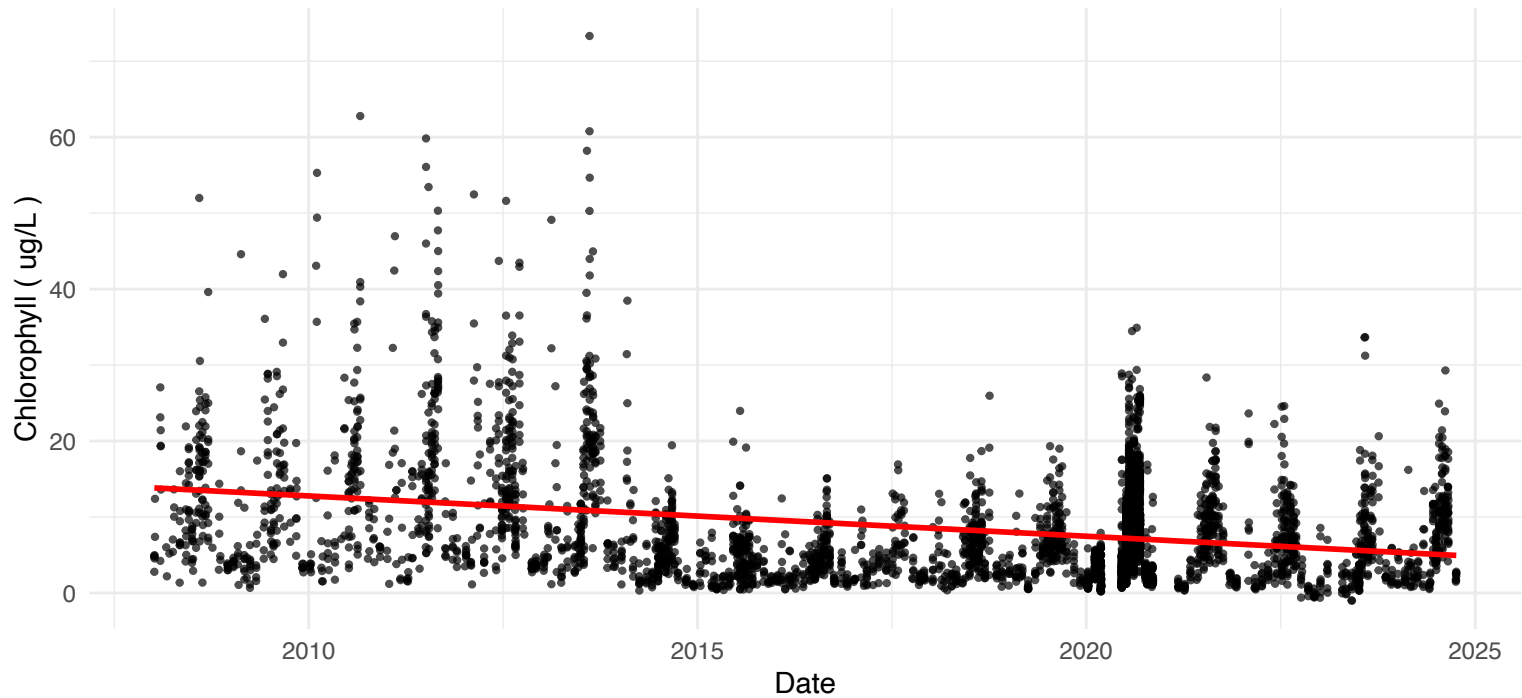
Redox potential from 2010–2024



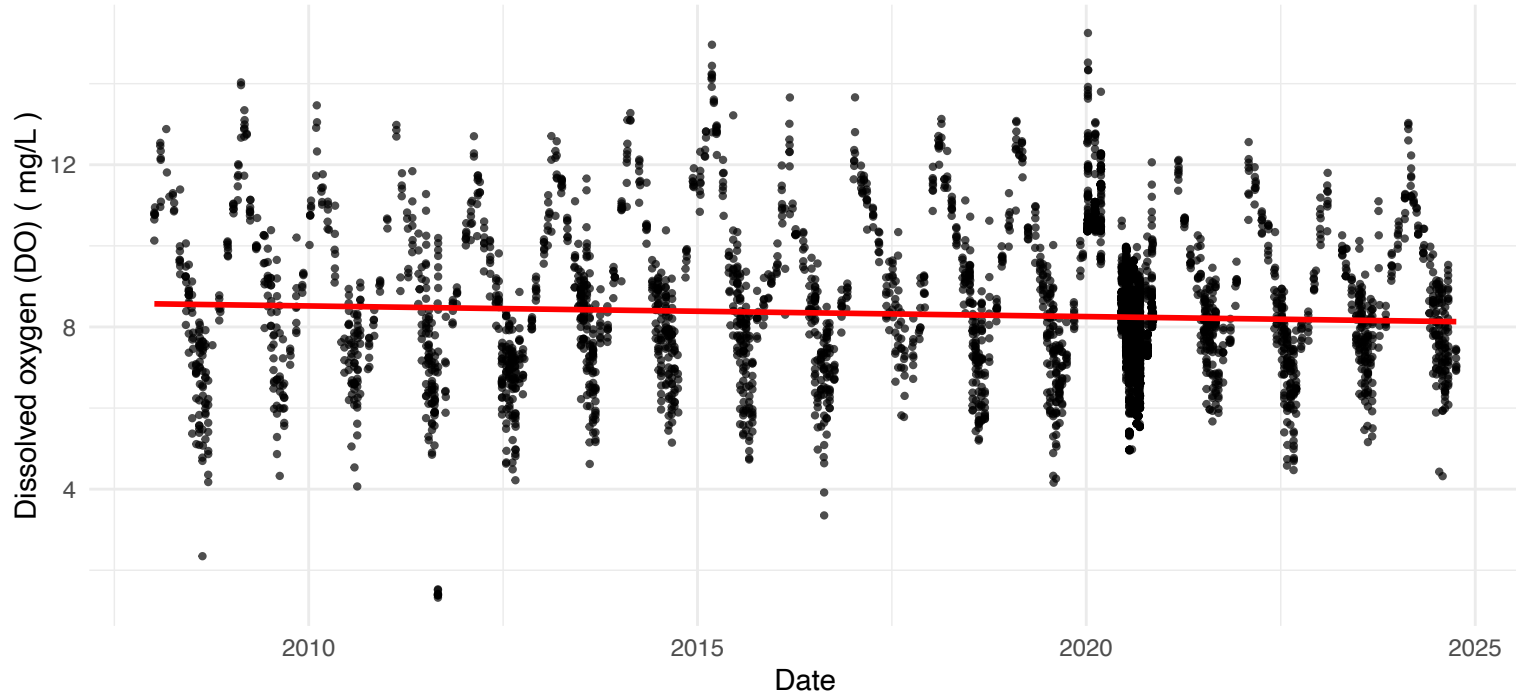
Redox potential from 2008-2024



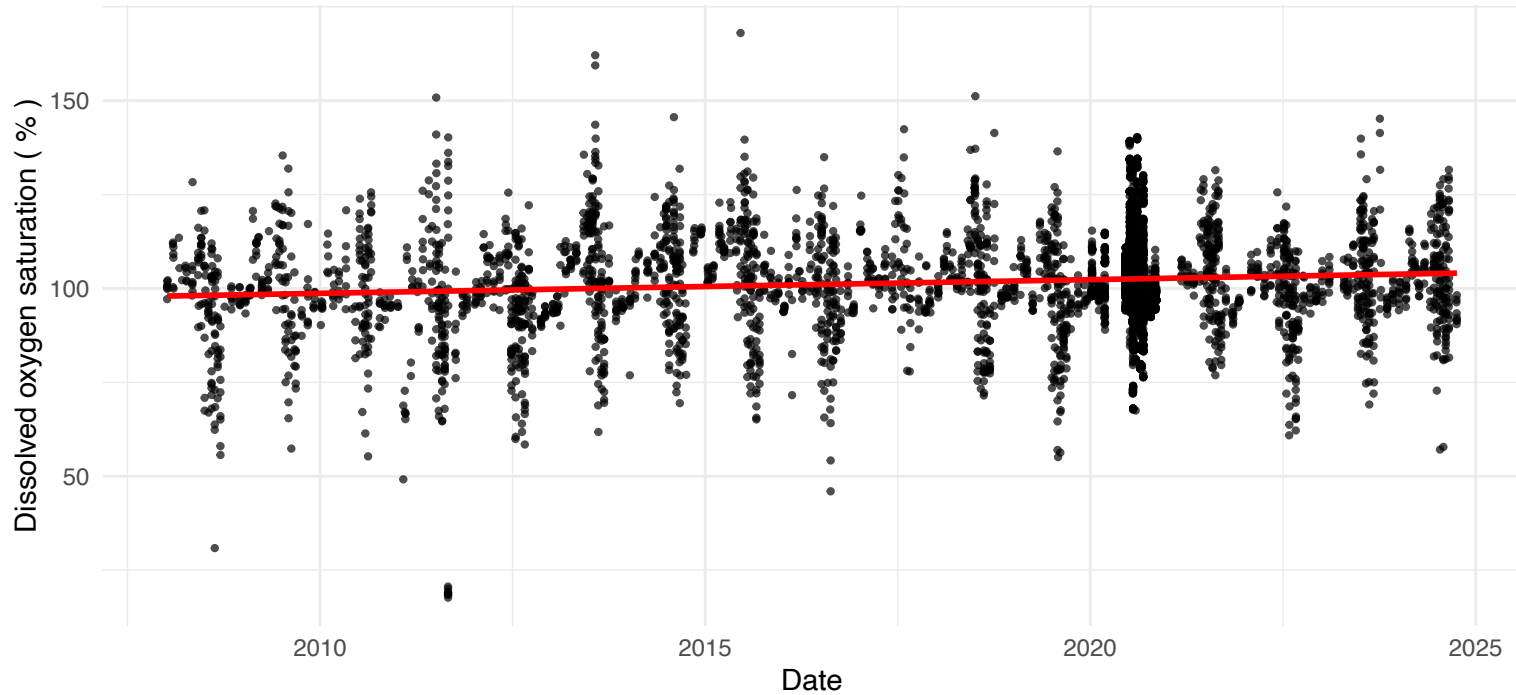
Chlorophyll from 2008–2024



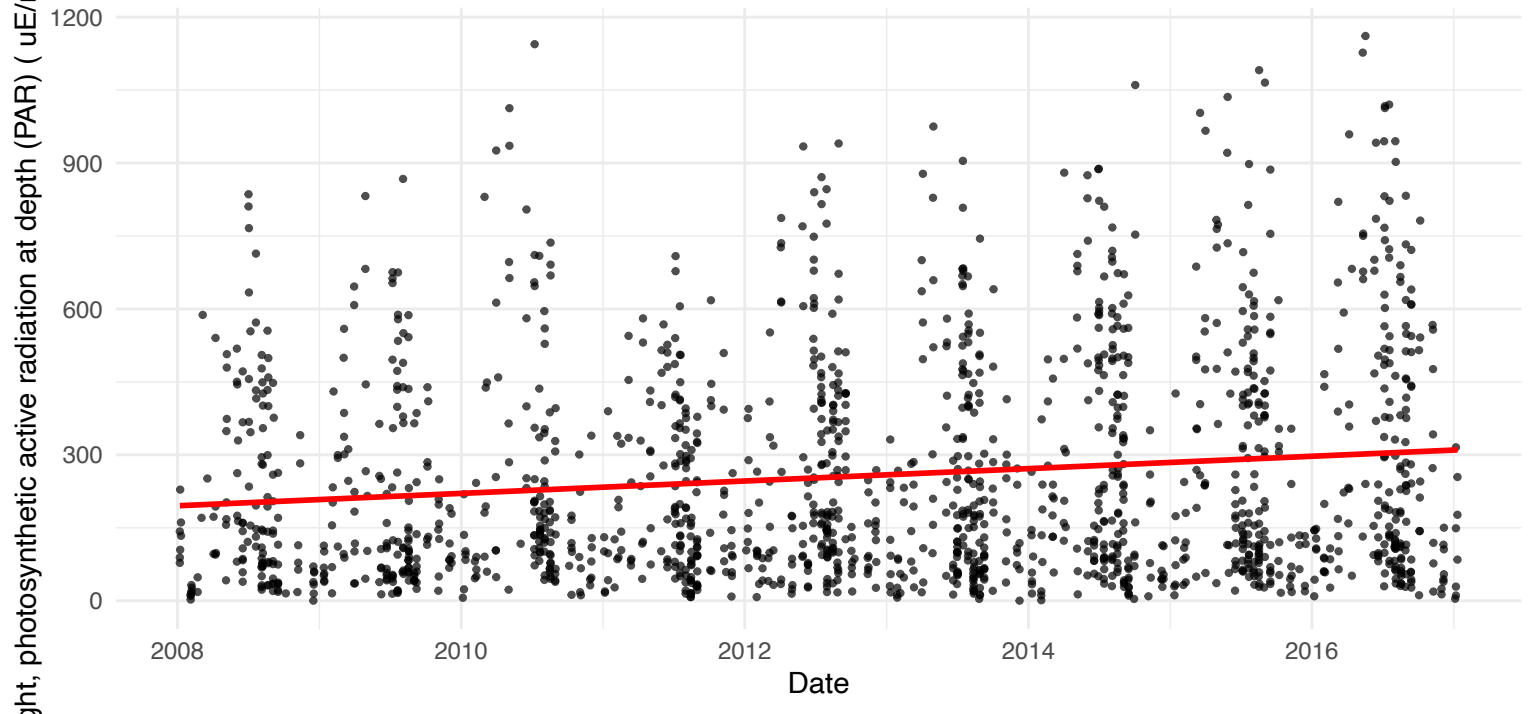
Dissolved oxygen (DO) from 2008–2024



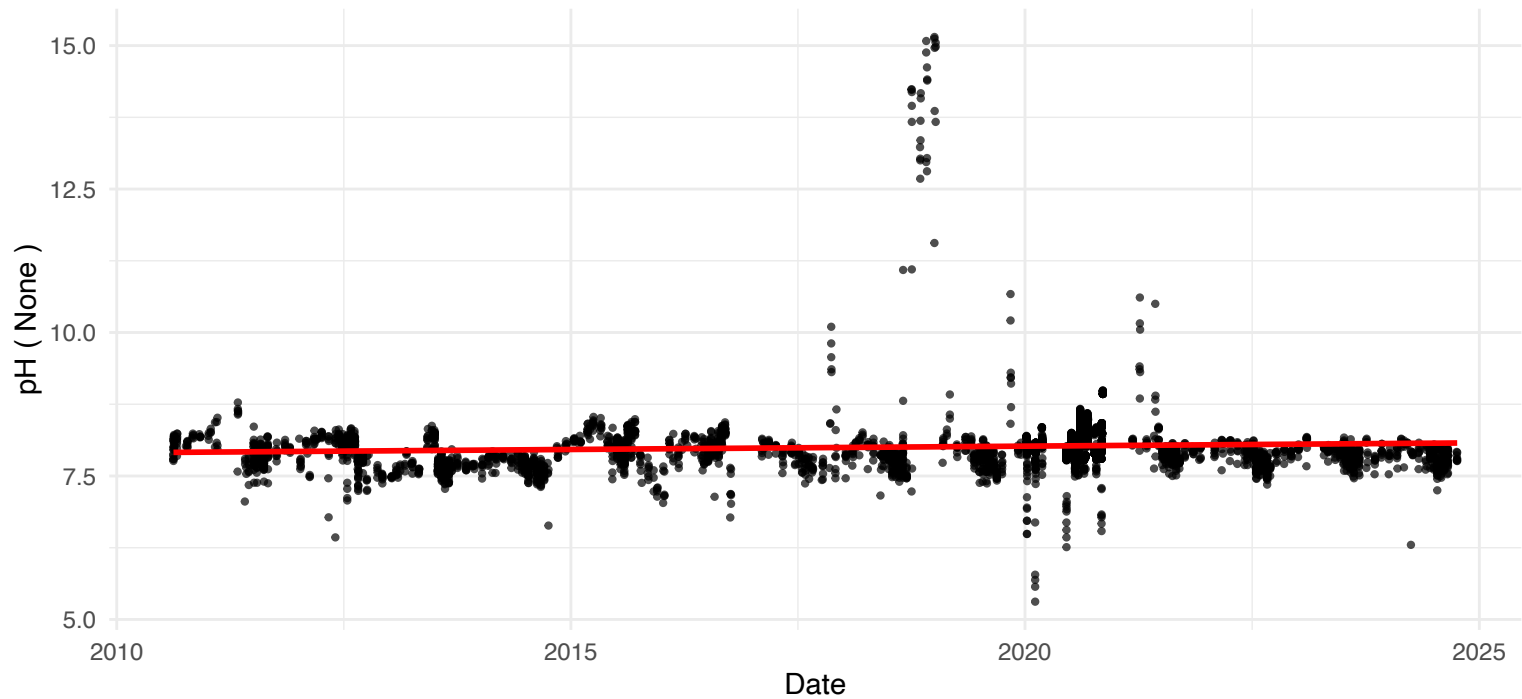
Dissolved oxygen saturation from 2008–2024



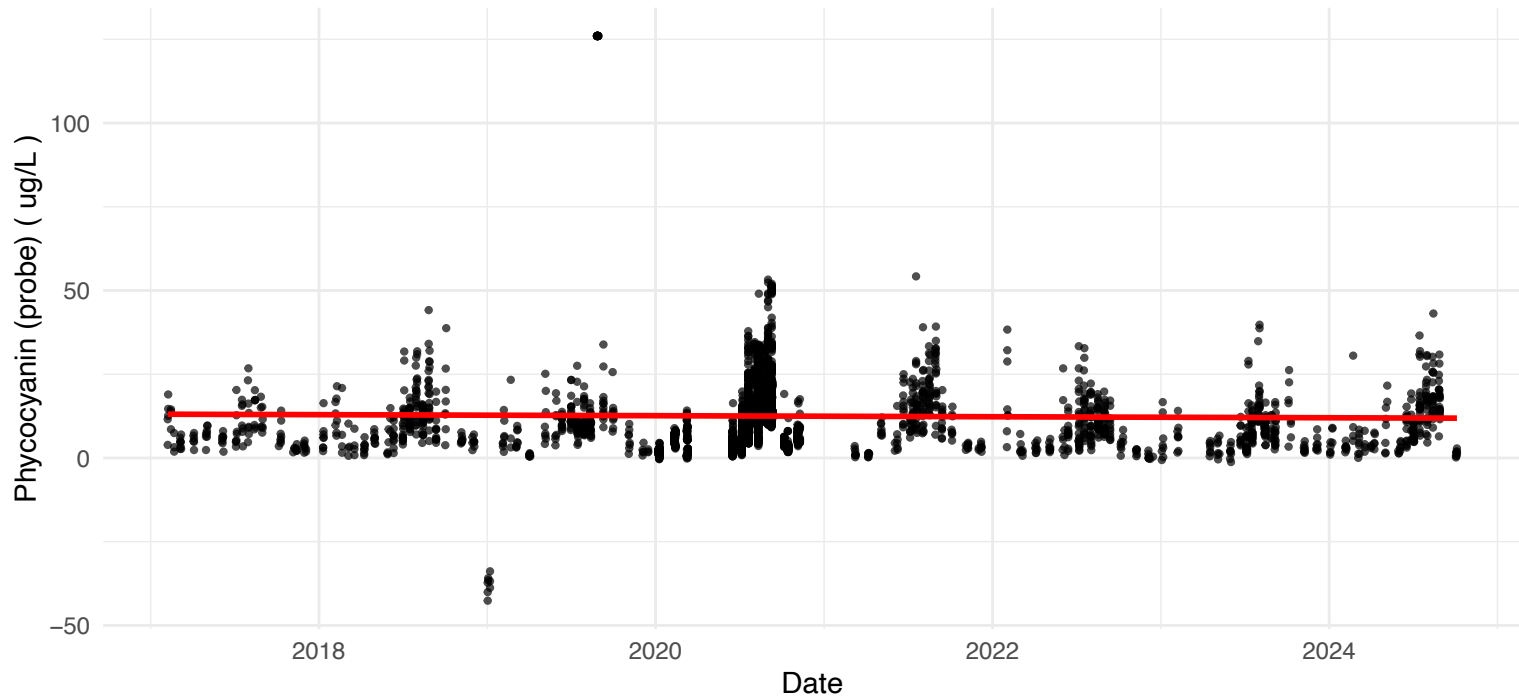
Light, photosynthetic active radiation at depth (PAR) from 2008–2024



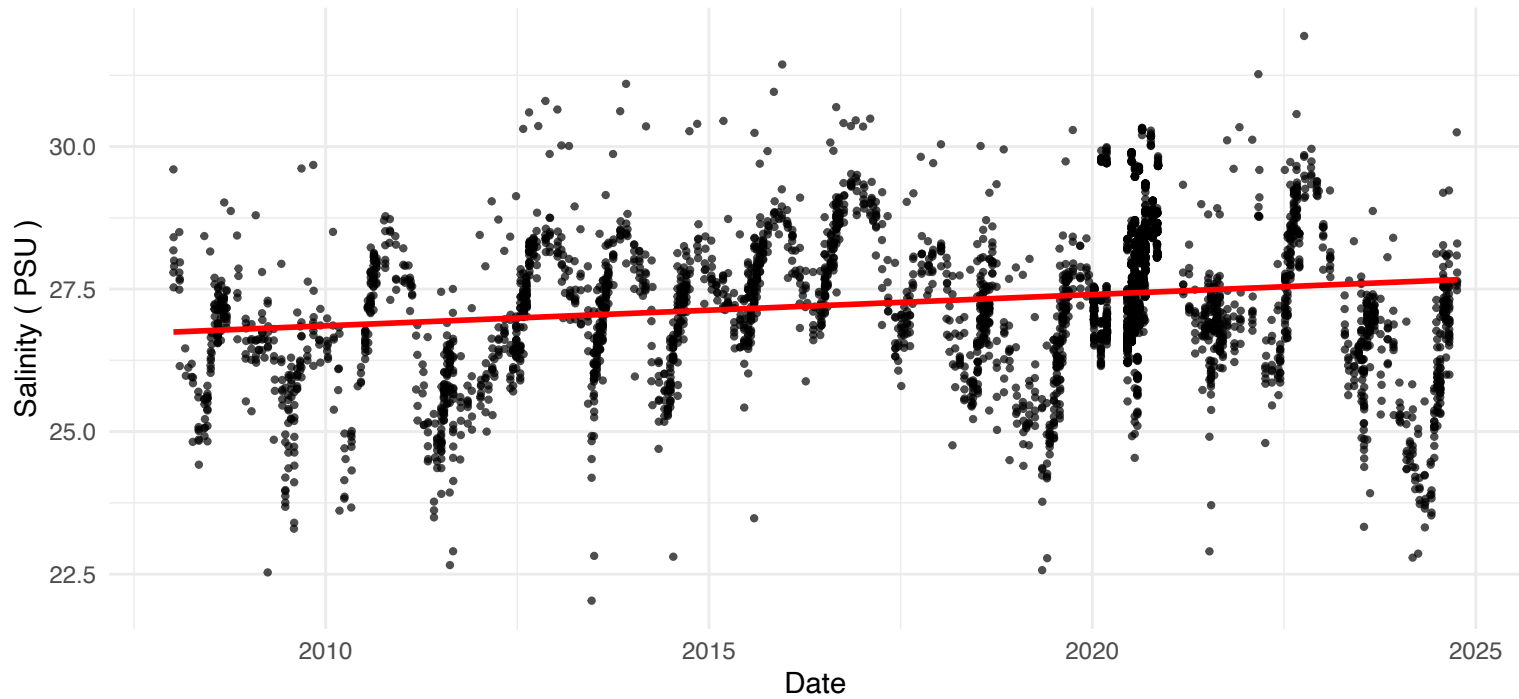
pH from 2008–2024



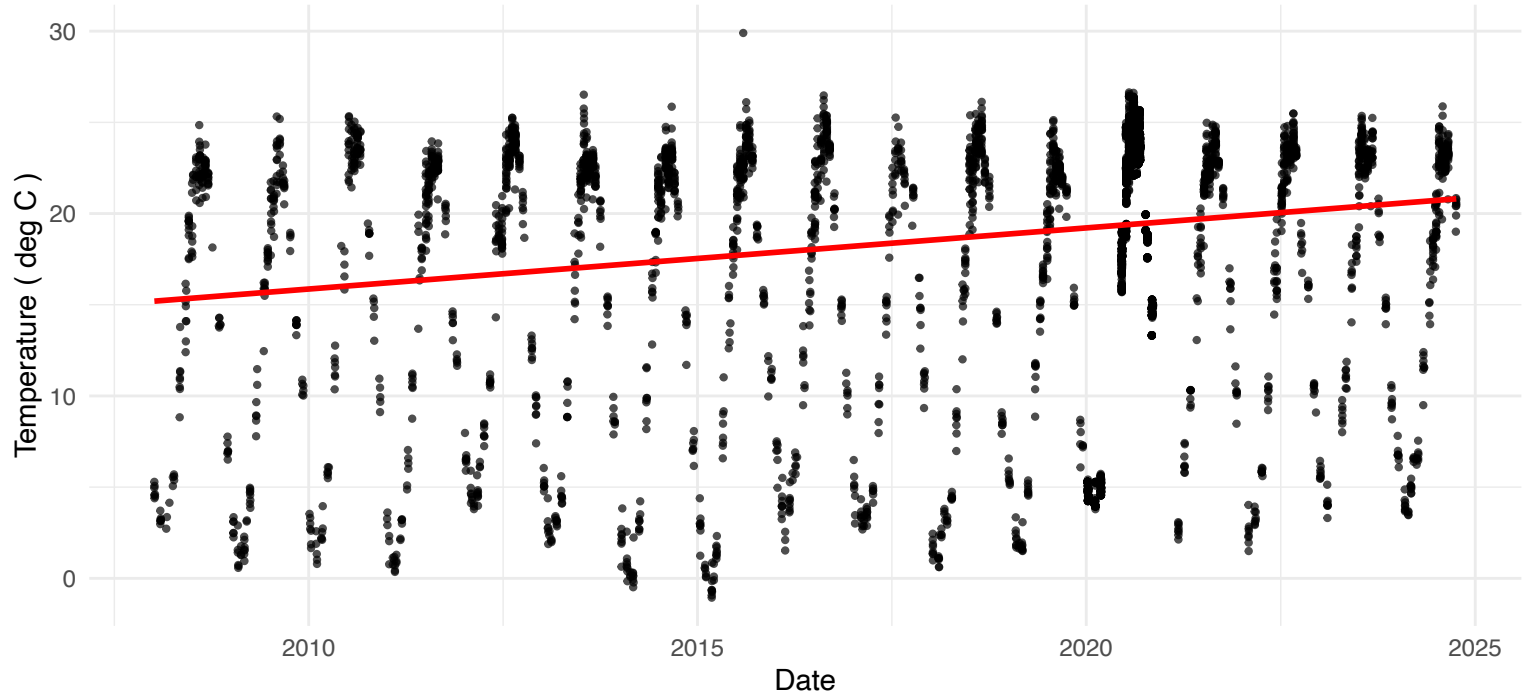
Phycocyanin (probe) from 2008–2024



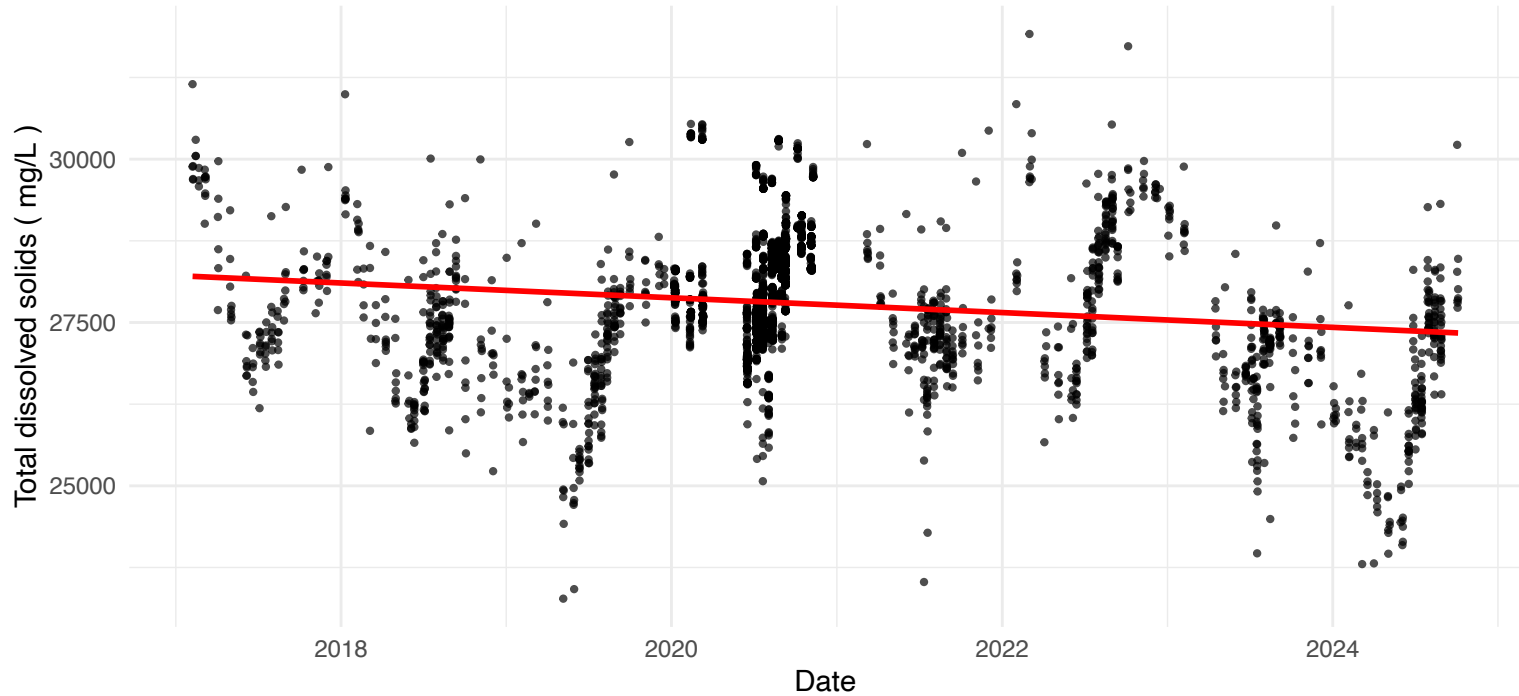
Salinity from 2008–2024



Temperature from 2008–2024



Total dissolved solids from 2008–2024



Turbidity from 2008–2024

